OligoTox-CNS — Public Access and Dissemination Plan

NIH/NCATS Oligonucleotide Toxicity Open Data Challenge — Phase 2 · CNS / neurotoxicity module · release v1.0

> **This plan is binding.** The Challenge states that award recipients must agree to abide by the terms of the PADP they submit, and that NIH intends to post it publicly. Everything below is written to be honoured as stated, not as an aspiration. Sections 3, 4 and 5 correspond one-to-one to the three requirements the Challenge sets for this document.

1 What is covered by this plan

"The solution" means the whole CNS module, not only its tables: the dataset (1,845 oligonucleotides, 4,394 CNS toxicity measurements, 32,569 per-position chemical-modification records), the build pipeline that produces it, the schema and data dictionary, the quality-control suite, the figures, and the narrative and methodology documents. A dataset without the code and vocabulary needed to use it is not usable, so the plan covers all of it together.

Component	Form	Covered
Dataset	CSV per endpoint, plus an XLSX workbook	yes
Data dictionary and schema	Markdown, generated from the code	yes
Build pipeline and QC suite	Python, no proprietary dependencies	yes
Source material	the retrieved files the pipeline reads, where redistributable	yes — see §6
Narrative and methodology documents	PDF	yes
Measurement instruments and grading rubric	Markdown	yes

2 Where it will live, and how people will find it

Channel	Purpose	Terms	When
Public git repository	canonical source; full history, so any figure or number traces to the commit that produced it	CC BY 4.0	live now
Zenodo deposit	archival copy with a citable DOI; a concept DOI always resolving to the latest version	CC BY 4.0	at submission
Challenge submission portal	the four required documents	as required by NIH	at submission

Zenodo is chosen over a domain-specific repository because no established public repository for oligonucleotide toxicity data exists — which is part of the gap this dataset addresses. There is **no registration, no data-use agreement, no access committee and no embargo**. Files are plain CSV, XLSX and Markdown, openable without proprietary software. No component requires a licence key, a hosted service, or a credential of any kind.

Dissemination of the knowledge to use it, not only the files. The Challenge asks for the knowledge necessary to access and utilise the dataset, so the repository ships the schema and data dictionary, the measurement instruments verbatim from their sources, the grading rubric with the rule recorded on every graded row, a worked baseline model that trains from the released files alone, and a build pipeline that regenerates every artefact from the committed sources. Announcement will be through a preprint describing the dataset, the Zenodo DOI, and direct notice to the oligonucleotide-safety groups whose published work the dataset curates.

3 Non-exclusive licensing for research purposes

Every licence granted under this plan is non-exclusive, irrevocable, worldwide and royalty-free. No exclusive licence to any part of the solution has been granted, and none will be. No party — including any future commercial partner — will receive rights that would prevent any researcher from using the dataset.

Component	Licence	What a researcher may do
Everything created by this project — schema, code, documentation, figures, all derived and computed fields	**CC BY 4.0**	any use, including commercial, with attribution
Rows curated from a CC BY source	CC BY 4.0	any use, with attribution to this dataset and the primary source
Rows curated from a CC BY-NC source	CC BY-NC 4.0	non-commercial research use, with attribution
Rows from US Government works (FDA labels)	public domain	any use
Build pipeline and QC suite	CC BY 4.0	any use, including commercial

A curated dataset cannot grant rights its sources did not grant, so terms are recorded **per row** in a `redistribution` column rather than asserted uniformly. 4,347 of 4,394 measurements (98.9%) are CC BY 4.0 or US public domain and are reusable for any purpose including commercially; the remainder derive from CC BY-NC sources, are individually marked, and are removable with a one-line filter. **This is stated rather than smoothed over: labelling the whole release CC BY would over-claim, and labelling it all non-commercial would needlessly restrict 98.9% of it.**

For research purposes specifically, no separate permission is needed for any part: the CC BY-NC portion already permits non-commercial research, and the remainder permits everything. A researcher can therefore use the entire dataset for research without contacting us.

4 If we cannot maximise public access ourselves

The Challenge requires a plan for the case where the winner is unable to maximise public access. Our answer is structural rather than promissory: **the licences above are already irrevocable, and the artefacts are already deposited in an archive we do not control.** A CC BY 4.0 grant cannot be withdrawn, so nothing we later do — or fail to do — can remove the public's rights to what has been released. The scenarios below therefore concern *continuity of stewardship*, not restoration of access.

Scenario	What happens	Licensing scheme employed
A. The team stops maintaining the dataset	The Zenodo deposit and its DOI persist independently of us. Anyone may fork, correct and redistribute the repository, including under their own name, provided attribution is kept.	CC BY 4.0 — already granted, irrevocable; no further act needed from us
B. A team member becomes unavailable	No single person holds a credential the release depends on. The repository has more than one maintainer, and the Zenodo record is deposited under an institutional-neutral account with a named co-depositor.	unchanged; stewardship transfers, licence does not
C. Hosting fails or a provider withdraws	Every source file the pipeline reads is committed alongside the code, so the dataset rebuilds from any surviving copy. Mirrors may be created by anyone without asking.	CC BY 4.0 permits redistribution and mirroring by any party
D. We are unable or unwilling to act on a correction	Any party may publish a corrected derivative. We will additionally grant, on written request and at no cost, a non-exclusive licence naming that party as a maintainer of record.	CC BY 4.0 for the derivative; a separate written non-exclusive stewardship licence
E. A third party wants terms beyond CC BY-NC on the NC-derived rows	We cannot grant what we do not hold. We will identify the upstream rightsholder and the exact rows affected so the requester can seek permission directly; the other 98.9% needs no permission.	referral, plus our own CC BY 4.0 grant on the structure, code and derived fields

5 Permitting the U.S. Government to enable others

The Challenge requires a plan describing how the winner will permit the U.S. Government to allow interested parties to utilise the solution, should the winner fail to utilise it and fail to permit others to do so on reasonable terms.

> **Standing grant.** We hereby grant the United States Government a non-exclusive, irrevocable, worldwide, royalty-free, fully paid-up licence to use, reproduce, distribute, prepare derivative works of, publicly display and publicly perform the solution, and **to authorise others to do so on its behalf**, for any purpose. This grant is made now and is not conditional on any failure by us — it does not need to be triggered, and there is nothing for the Government to determine before exercising it.

Why the grant is unconditional

A conditional grant would require someone to adjudicate whether the winner had in fact failed to act, and on what timescale — which is precisely the delay the requirement exists to prevent. Granting it up front removes the determination step. In practice the grant is also redundant for the CC BY 4.0 portion, which already permits the Government and everyone else to do all of the above; it matters for the CC BY-NC-derived rows, where it removes the non-commercial restriction as far as we are able to remove it.

Scenario	Government action available	Licensing scheme
We fail to maintain or disseminate the solution	Host, mirror, update or republish it directly, or authorise any third party to	the standing grant above; no notice to us required
We refuse a reasonable request from a third party	Authorise that party directly under the standing grant	standing grant; we waive any right to object
Government wishes to fund a derivative or extension	Commission it from any party, including a competitor	standing grant, plus CC BY 4.0 on the underlying work
Upstream CC BY-NC terms limit a Government use	We will identify the rightsholder and the affected rows and support the request	referral; our own rights already granted in full

The one limit we state honestly: we cannot grant rights in third-party copyrighted material we merely curated. That limit is bounded and documented row by row, and it touches 47 of 4,394 measurements.

6 Limits, stated plainly

- **Three source documents are deliberately not redistributed.** Two conference presentations and one journal correspondence item are copyrighted and not licensed for redistribution. They are named in the source register with everything needed to obtain them independently, and **no released row depends on any of them**. Curating a source is not the same act as republishing it, and we have not conflated the two.
- **The dataset contains no human *in vitro* data.** Of 4,394 measurements, 2341 are human clinical and **0 are human *in vitro***; the rest are animal. This plan is framed by the Challenge around "winning datasets from in vitro human-based systems", so the limit is material and we state it here rather than only in the narrative. The dataset's contribution is the sequence-to-toxicity pairing and the in-vitro-to-in-vivo structure; the human arm of that structure is not yet present.
- **Severity grades are provisional** pending subject-matter-expert review. Every grade carries the rule that produced it, so a reviewer can audit rather than re-derive.
- **No personal data is involved.** Every measurement is either a preclinical result or an aggregate incidence already published in a regulatory document. There are no human subjects, no identifiable information, and therefore no consent or privacy constraint on any part of the release.

7 Maintenance and preservation

Corrections are made by pull request against the public repository, with the 34-check quality suite as the merge gate. Each release is re-deposited to Zenodo with a new version DOI while the concept DOI continues to resolve to the latest. The repository is self-contained — the source files the pipeline reads are committed alongside the code — so the dataset remains rebuildable even if a publisher URL rots. Preservation does not depend on us: the Zenodo deposit is independently archived, and the CC BY 4.0 grant permits anyone to mirror it.